

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering



Project Report on Interview IQ: AI Interview System

**In partial fulfilment of the Fourth Year, Bachelor of Engineering (B.E.)
Degree in Computer Engineering at the University of Mumbai
Academic Year 2025 - 26**

By

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(A.Y. 2025 - 26)

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Certificate

This is to certify that **Toshit Hole (D17C-28), Shreyas Bagwe (D17C-05), Jatin Hargunani (D17C-27), Subodh Thalkari (D17C-62)** of Fourth Year Computer Engineering studying under the University of Mumbai have satisfactorily completed the project on “**Interview IQ: AI Interview System**” as a part of their coursework of PROJECT - II for Semester - VIII under the guidance of their mentor **Mrs. Vidya S Zope** in the year 2025 - 26.

This thesis/dissertation/project report entitled “**Interview IQ: AI Interview System**” by **Toshit Hole, Shreyas Bagwe, Jatin Hargunani & Subodh Thalkari** is approved for the degree of **Bachelor of Engineering in Computer Engineering**.

Programme Outcomes	Grade
PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12 PSO1 & PSO2	

Date:

Project Guide: Mrs. Vidya S Zope

Project Report Approval

For

B. E (Computer Engineering)

This project report entitled “**Interview IQ: AI Interview System**” by **Toshit Hole (D17C-28), Shreyas Bagwe (D17C-05), Jatin Hargunani (D17C-27), Subodh Thalkari (D17C-62)** is approved for the degree of **Bachelor of Engineering in Computer Engineering**.

Examiners

1.

(Internal Examiner name & sign)

2.

(External Examiner name & sign)

3.

(Head of Department)

4.

(Principal)

Date:

Place: Chembur, Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

COURSE OUTCOMES FOR B.E. PROJECT

Learners will be able to,

Course Outcome	Description of the Course Outcome
CO1	Able to apply the relevant engineering concepts, knowledge and skills towards the project.
CO2	Able to identify, formulate and interpret the various relevant research papers and to determine the problem.
CO3	Able to apply the engineering concepts towards designing solutions for the problem.
CO4	Able to interpret the data and datasets to be utilized.
CO5	Able to create, select and apply appropriate technologies, techniques, resources and tools for the project.
CO6	Able to apply ethical, professional policies and principles towards societal, environmental, safety and cultural benefit.
CO7	Able to function effectively as an individual, and as a member of a team, allocating roles with clear lines of responsibility and accountability.
CO8	Able to write effective reports, design documents and make effective presentations.
CO9	Able to apply engineering and management principles to the project as a team member.
CO10	Able to apply the project domain knowledge to sharpen one's competency.
CO11	Able to develop a professional, presentational, balanced and structured approach towards project development.
CO12	Able to adopt skills, languages, environment and platforms for creating innovative solutions for the project.

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Abstract

InterviewIQ – An AI-Powered Mock Interview Platform is an advanced and intelligent system designed to revolutionize the way individuals prepare for interviews by integrating artificial intelligence, natural language processing, and multimodal analysis. The platform leverages multiple Large Language Models (LLMs) to simulate highly realistic and adaptive interview experiences, providing users with an environment that mirrors real-world professional interactions. It offers both audio and video interview modes, where the audio module focuses on evaluating verbal attributes such as tone, fluency, filler word usage, long pauses, and overall speech confidence, while the video module—powered by OpenCV and MediaPipe—analyzes non-verbal communication aspects such as eye contact, gestures, facial expressions, and posture. A unique feature of InterviewIQ is its AI Interviewer Personalization System, allowing users to choose from four distinct virtual interviewer personas, each with different temperaments and questioning approaches, thereby offering a psychologically diverse and engaging experience.

The platform is designed to provide in-depth analytics and performance feedback post-interview, including verbal and behavioral insights, confidence scores, and detailed evaluations highlighting strengths and areas for improvement. Furthermore, InterviewIQ integrates real-time socket-based communication for seamless interaction, low-latency response generation, and advanced speech-to-text synthesis for accurate evaluation. It also plans to introduce a Retrieval-Augmented Generation (RAG)-based feature, enabling users to upload personal documents or notes from which the AI interviewer can generate context-specific questions and assess responses accordingly.

By combining the capabilities of multi-agent orchestration, cross-LLM intelligence, and adaptive multimodal evaluation, InterviewIQ offers a highly interactive, data-driven, and user-centric approach to interview preparation, empowering individuals to enhance their communication, confidence, and analytical skills, while bridging the gap between traditional mock interviews and next-generation AI-driven career readiness solutions.

Chapter 1: Introduction

1.1 Introduction

Interview preparation plays a pivotal role in shaping an individual's professional journey, as it directly impacts employability, communication confidence, and overall career readiness. Despite the abundance of online learning resources, most candidates still struggle to find platforms that offer personalized, realistic, and adaptive practice environments. Conventional approaches—such as question banks, video tutorials, or manual mock sessions—fail to capture the dynamic and interactive nature of real interviews. These methods often lack flexibility, are costly, and provide minimal feedback on both verbal and non-verbal performance parameters.

With the rapid progress in Artificial Intelligence (AI) and Machine Learning (ML), intelligent interview simulation has emerged as a transformative solution for bridging this gap. AI-driven systems can simulate human-like interactions, generate adaptive questions, and evaluate candidate performance using speech, facial, and behavioral cues. However, most existing tools remain limited to text-based interfaces, offering little real-time adaptability or multimodal analysis. They often fail to assess communication confidence, body language, or engagement, which are crucial for holistic interview preparation.

InterviewIQ addresses these limitations by introducing a multi-agent, multimodal AI-powered mock interview platform that enables users to practice interviews through both audio and video modes. The system utilizes powerful LLM-based intelligence to create realistic, domain-specific, and context-aware interview experiences. It further evaluates users on speech patterns, filler words, gesture tracking, and confidence levels—providing detailed analytical feedback and performance reports. By combining adaptability, inclusivity, and interactivity, InterviewIQ aims to redefine how candidates prepare for real-world interviews.

1.2 Motivation

In today's competitive job landscape, candidates face immense pressure during interviews, often due to lack of realistic practice and constructive feedback. While

professional coaching and paid mock interviews exist, they are expensive, geographically restricted, and non-scalable. Furthermore, the majority of existing online tools are limited to text-based question-answer modules, offering no real-time adaptability or behavioral assessment.

The motivation behind developing InterviewIQ stems from the need to democratize high-quality interview preparation through AI-driven intelligence. The system not only helps users improve their technical and communication skills but also focuses on confidence building by simulating real-life interaction scenarios. It offers multiple interview types such as Technical, HR, Finance, Marketing, and Sales, with subcategories like Data Structures and Algorithms, Frontend, Backend, and DevOps for technical interviews. This allows users to prepare for specific domains of their choice.

A distinctive motivation is to create diverse AI interviewer personas, enabling users to choose their interviewer's gender, tone, and attitude. The system includes four interviewer profiles—two female (a strict senior interviewer and a confident young professional) and two male (an experienced elder mentor and a young supportive interviewer). These personalities help simulate varied interview environments, reducing anxiety and improving adaptability. Thus, InterviewIQ's motivation lies in creating a personalized, affordable, and psychologically enriching learning experience for every user.

1.3 Problem Definition

Despite the technological progress in AI-based assessment tools, interview simulation remains underdeveloped in terms of realism, personalization, and feedback. Existing systems suffer from the following limitations:

- **Lack of Adaptability:** Most platforms use pre-defined question sets without context-aware follow-ups.
- **Absence of Multimodal Analysis:** Current solutions rarely combine audio and video cues for holistic evaluation.
- **Limited Diversity:** Users cannot choose interviewer behavior or tone, which is crucial for real-world readiness.

- **Inadequate Feedback:** Feedback is often generic and lacks detailed analytics on communication or behavioral traits.
- **Cost and Accessibility Issues:** Many tools rely on premium APIs or closed-source platforms, restricting access.

Hence, the core problem addressed by this project is: To design and develop a scalable, multimodal AI-powered interview simulation platform that integrates multiple LLMs and computer vision techniques to evaluate users comprehensively across technical, communication, and behavioral parameters, while offering interviewer personalization and detailed analytical feedback.

1.4 Relevance of the Project

InterviewIQ holds significant relevance in the modern employment and education ecosystem, where virtual hiring and remote assessments are rapidly becoming the norm. Organizations are increasingly adopting AI-assisted recruitment processes, and candidates must adapt to these evolving interview environments. By providing a realistic, adaptive, and feedback-oriented mock interview platform, InterviewIQ aligns with global trends in AI-based education and professional training.

From an academic and technological perspective, the project contributes to the domains of Conversational AI, Computer Vision, and Speech Processing, offering practical insights into their integration. It supports the goals of AI-driven skill development and digital employability enhancement. Furthermore, the project aligns with the United Nations Sustainable Development Goal (SDG) 8 – Decent Work and Economic Growth, by enabling equitable access to career-preparation resources.

In summary, InterviewIQ's relevance is defined by its potential to:

- Enhance confidence and communication skills through realistic simulation.
- Bridge the accessibility gap by offering an affordable, cloud-based platform.
- Enable fair and transparent evaluation using data-driven analytics.
- Advance AI's role in ethical, inclusive, and personalized education.

1.5 Methodology Used

The development methodology for InterviewIQ is structured into distinct but interconnected stages that ensure accuracy, adaptability, and performance efficiency:

Step 1: Data and Model Integration

The platform integrates multiple LLMs (OpenAI, Claude, Gemini, Groq) for diverse question generation, response evaluation, and contextual adaptation. It leverages publicly available question datasets and fine-tuned dialogue prompts to ensure domain variety and natural flow.

Step 2: Multimodal Architecture Development

Two interaction modes—Audio and Video—are implemented. The Audio module utilizes Whisper ASR for speech-to-text conversion and Coqui TTS for AI interviewer voice synthesis. The Video module employs OpenCV and MediaPipe to detect facial expressions, eye contact, and gestures. Combined, these create a holistic feedback mechanism covering both verbal and non-verbal communication.

Step 3: Interviewer Personalization

A multi-agent architecture assigns unique behavioral parameters to each AI interviewer persona. This affects question tone, difficulty, and interaction style, allowing users to experience varied psychological environments similar to real interviews.

Step 4: Analytics and Feedback System

Post-interview, the platform generates a comprehensive analytical report summarizing metrics like filler word frequency, pause duration, confidence score, facial engagement, and response relevance. These are visualized using dashboards for clarity and progress tracking.

Step 5: Deployment and Testing

The entire system is built using a MERN-based stack (MongoDB, Express, React, Node.js) with Python microservices for AI modules. The web application will be deployed on cloud infrastructure, ensuring accessibility and scalability. User feedback and continuous testing will guide refinements for Stage-2 development.

Chapter 2: Literature Survey

2.1 Research Papers Referred

Sr no	Paper Title & Authors	Year	Description / Methodology	Inference / Relevance to Project
1	Chen, L., Wang, S., & Liu, H. – "AI in Education: Applications and Methodologies"	2023	Discusses the application of AI models in educational systems for adaptive learning and student evaluation.	Establishes how AI can personalize assessment processes and improve learner outcomes.
2	Orak, C. & Turan, Z. – "Using AI in Digital Interview and Assessment Systems"	2024	Explores the use of NLP and sentiment analysis in virtual interviews to measure communication skills.	Provides insight into using NLP for candidate evaluation, applicable to InterviewIQ's feedback module.
3	Patel, D., & Mehta, V. – "Speech Recognition for Interactive Learning Environments"	2023	Describes speech recognition (ASR) integration for interactive educational tools.	Supports InterviewIQ's use of Whisper ASR for speech-to-text conversion.
4	Gupta, P., Sharma, K., & Rao, A. – "Machine Learning Models for Soft Skill Assessment"	2022	Describes ML models for evaluating communication and personality traits.	Reinforces the need for analytics-driven evaluation of communication and personality traits.
5	Krishnan, N. – "AI Agents: Architecture and Real-World Applications"	2025	Explains how multi-agent systems coordinate tasks in complex AI applications.	Provides a foundation for InterviewIQ's multi-agent orchestration design.

2.2 Patent Search Summary

Sr. No.	Patent Title	Patent No. / Year	Relevance to Interview-IQ
1	System and Method for Conducting Virtual Interviews Using Artificial Intelligence	US 2021015678 9 A1 (2021)	Provides foundational insight into integrating AI for candidate evaluation and emotion recognition during interviews.
2	Interactive Interview Simulation Platform with Speech and Emotion Recognition	EP 3856721 A2 (2022)	Supports the development of InterviewIQ's audio-interview mode using speech analytics and filler word detection.
3	Method for Generating Dynamic Question Sets in AI-Based Training Systems	US 2022009145 7 A1 (2022)	Aligns with InterviewIQ's adaptive question-generation system powered by LLMs for context-aware questioning.
4	Computer Vision-Based Behavioral Assessment System	WO 2020034781 A1 (2020)	Forms the conceptual basis for InterviewIQ's video interview mode using OpenCV and

			MediaPipe for gesture and eye-contact analysis.
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2.3 Existing Systems

Tool Name	Key Features	Target Users	Limitations
Final Round AI	Provides real-time AI assistance during live interviews; supports answer structuring, coding help, and JD-based preparation.	Job seekers needing real-time guidance during technical or HR interviews.	May raise ethical concerns if used during actual interviews; limited learning feedback.
Remasto AI	Offers mock interviews based on job roles or companies; includes resume builder and basic feedback analytics.	Students and professionals preparing for role-specific interviews.	Text-based only; lacks voice interaction and adaptive learning capability.
Interviews Chat	Conducts AI-based mock interviews with smart follow-up questions and performance summaries; includes resume and cover letter review.	Candidates preparing for behavioral and technical interviews.	No voice or video-based interaction; feedback is generic and not deeply personalized.
Sapia AI	Enterprise tool for AI-driven candidate screening; performs text-based, bias-free assessments with personality analytics.	HR teams and recruiters conducting large-scale hiring.	Designed for companies, not learners; lacks real-time or speech-based interaction.

2.4 Lacuna in Existing Systems

While existing systems demonstrate the growing influence of AI in interview preparation and recruitment, several gaps remain unaddressed:

1. **Absence of Speech Interaction:** Most tools rely solely on text-based inputs without incorporating natural speech recognition or voice synthesis.
2. **Lack of Real-Time Adaptability:** Questions are pre-defined and do not evolve based on user responses.
3. **Limited Educational Feedback:** Current platforms focus on evaluation rather than continuous learning improvement.

2.5 Comparison of Existing System and Proposed Work

Sr no	Existing Systems	Proposed Work
1	Existing platforms primarily offer text-based or limited interactive mock interview experiences. These systems generate pre-defined questions without real-time adaptability and often lack multimodal input support.	InterviewIQ provides fully interactive audio and video-based interviews powered by LLMs capable of adapting questions dynamically to user responses, ensuring a realistic and personalized experience.
2	Most existing tools focus only on textual responses and ignore non-verbal communication elements such as tone, pauses, or gestures.	InterviewIQ integrates speech analysis (detecting filler words, pauses, and tone) and computer vision analytics (eye contact, facial expressions, gestures) for holistic evaluation.
3	Feedback provided in existing systems is generic or score-based, offering limited actionable insights.	InterviewIQ generates comprehensive performance reports with analytics on communication clarity, confidence, domain knowledge, and behavioral traits.
4	Existing solutions do not allow interviewer personality customization and treat every interaction the same way.	InterviewIQ introduces AI interviewer personas—four distinct profiles (two male, two female) with varying tones and strictness, allowing users to choose interviewer preferences.

2.6 Focused Work

The primary focus area of InterviewIQ is to develop a multimodal, adaptive, and intelligent interview simulation platform that provides a realistic and personalized experience to every user. The system emphasizes AI-driven question adaptability, speech and gesture-based evaluation, and customizable interviewer personas to simulate diverse real-world interview environments.

By integrating advanced LLM-based reasoning, speech analytics, and computer vision modules, InterviewIQ ensures holistic assessment across technical, communication, and behavioral dimensions. The project's central aim is to make interview preparation interactive, data-driven, and accessible, enabling users to gain confidence, receive constructive insights, and enhance their employability through continuous AI-guided improvement.

Chapter 3: Requirement of Proposed System

3.1 Proposed Model

The proposed model of InterviewIQ follows a modular, service-oriented architecture designed for scalability, adaptability, and real-time performance. It is structured into three core layers — the Presentation Layer, Application Layer, and Data & AI Layer. The Presentation Layer, built using React.js and Tailwind CSS, serves as the user interface where candidates interact through audio or video interviews. The Application Layer, developed with Node.js and Express, manages communication between the frontend and backend through secure RESTful APIs and WebSockets for real-time data flow.

The AI and Data Layer integrate multiple components, including LLM-based question generation, speech analysis (Whisper ASR), and computer vision modules (OpenCV & MediaPipe). These components collectively handle natural language understanding, filler word detection, facial and gesture analysis, and feedback synthesis. All data—such as user profiles, session records, and analytics reports—are securely stored in MongoDB. This layered architecture ensures modularity, allowing each subsystem (AI, video processing, analytics, and storage) to function independently while maintaining seamless integration across the platform.

3.2 Functional Requirements

The functional requirements of InterviewIQ define the core operations and behaviors expected from the system to ensure smooth and intelligent interview simulation. Key Functional Requirements:

- 1. User Registration and Authentication** – The system must allow users to securely register, log in, and manage profiles.
- 2. Interview Mode Selection** – Users should be able to choose between audio and video interview modes.
- 3. Domain and Subdomain Selection** – Users must select the interview category (e.g., Technical, Finance, Marketing, Sales) and subdomain (e.g., DSA, Frontend, Backend).

- 4. AI Interviewer Personalization** – The system should enable users to select from four AI interviewer personas (strict/lenient, male/female).
- 5. Dynamic Question Generation** – The AI module should generate context-aware, adaptive questions using LLMs based on user responses.
- 6. Speech Processing** – In audio interviews, the system must analyze speech for filler words, long pauses, tone, and fluency.
- 7. Video Analysis** – In video interviews, the system should use OpenCV and MediaPipe to evaluate eye contact, gestures, and facial expressions.
- 8. Real-time Feedback** – The system should provide real-time conversational flow between the AI interviewer and the user.
- 9. Performance Analytics Generation** – After each interview, the system must generate a detailed report showing scores for technical accuracy, communication, confidence, and behavioral indicators.
- 10. Interview History Management** – The platform must store all past interview sessions and analytics for future reference.
- 11. Admin Dashboard** – An admin module should monitor usage statistics, feedback trends, and system performance.
- 12. Cloud Deployment and Accessibility** – The system should be deployable on cloud servers, allowing access from any device through a web browser.

3.3 Non-Functional Requirements

Performance: Real-time AI processing must be efficient, though slight latency may occur.

Usability (UX): Must feature a user-friendly interface.

Reliability: The system must ensure accurate speech recognition (Whisper ASR) and realistic voice synthesis (Coqui TTS).

Scalability: Must be able to handle increasing user traffic and data.

Security: Implement authentication middleware and API request validation for secure operations.

3.4 Hardware & Software Requirements

Hardware Requirements:

- Development System: Laptop/PC with minimum Intel i5 / AMD Ryzen 5 processor or higher.
- Memory (RAM): Minimum 8 GB RAM (16 GB recommended for smoother AI and video processing).
- Storage: Minimum 256 GB SSD for local development and dataset storage.
- Graphics Support: GPU-enabled system (optional but beneficial for faster AI inference).
- Internet Connectivity: Stable broadband connection for accessing cloud services and APIs.
- Deployment Server: Cloud-based virtual machine (AWS EC2 / Google Cloud / Render) with 4 vCPUs, 8 GB RAM, and Docker support.

Software Requirements:

- Operating System: Windows 10/11, macOS, or Ubuntu Linux.
- Frontend: React.js, Tailwind CSS.
- Backend: Node.js, Express.js.
- Database: MongoDB (v5.0 or later).
- AI and ML Modules: Python (v3.10+), LangChain, Whisper ASR, Coqui TTS, OpenCV, MediaPipe.
- API and Development Tools: Postman, Docker & Docker Compose, Git & GitHub, VS Code.
- Web Communication Frameworks: WebRTC, Socket.io for real-time interview interactions.
- Cloud Deployment Tools: NGINX (reverse proxy and static file serving), Render / Vercel for hosting.

3.5 Constraints of Working System

- **API Dependency:** The system's performance depends on third-party AI APIs for language, speech, and vision processing, which may introduce latency or downtime.
- **High Computational Demand:** Real-time audio and video processing, along with LLM inference, require substantial processing power and memory resources.
- **Network Reliability:** The platform's real-time interaction relies heavily on stable internet connectivity; poor bandwidth can disrupt interviews or delay responses.
- **Operational Cost:** Using premium APIs and cloud-based GPU instances increases development and maintenance costs.
- **Data Privacy and Security:** Storing and processing user audio/video data must comply with strict security standards to ensure confidentiality and prevent misuse.
- **Limited Time for Development:** Academic project timelines restrict the integration of advanced features and extensive user testing within the initial development phase.

Chapter 4: Proposed Design

4.1 Block Diagram of the Proposed System

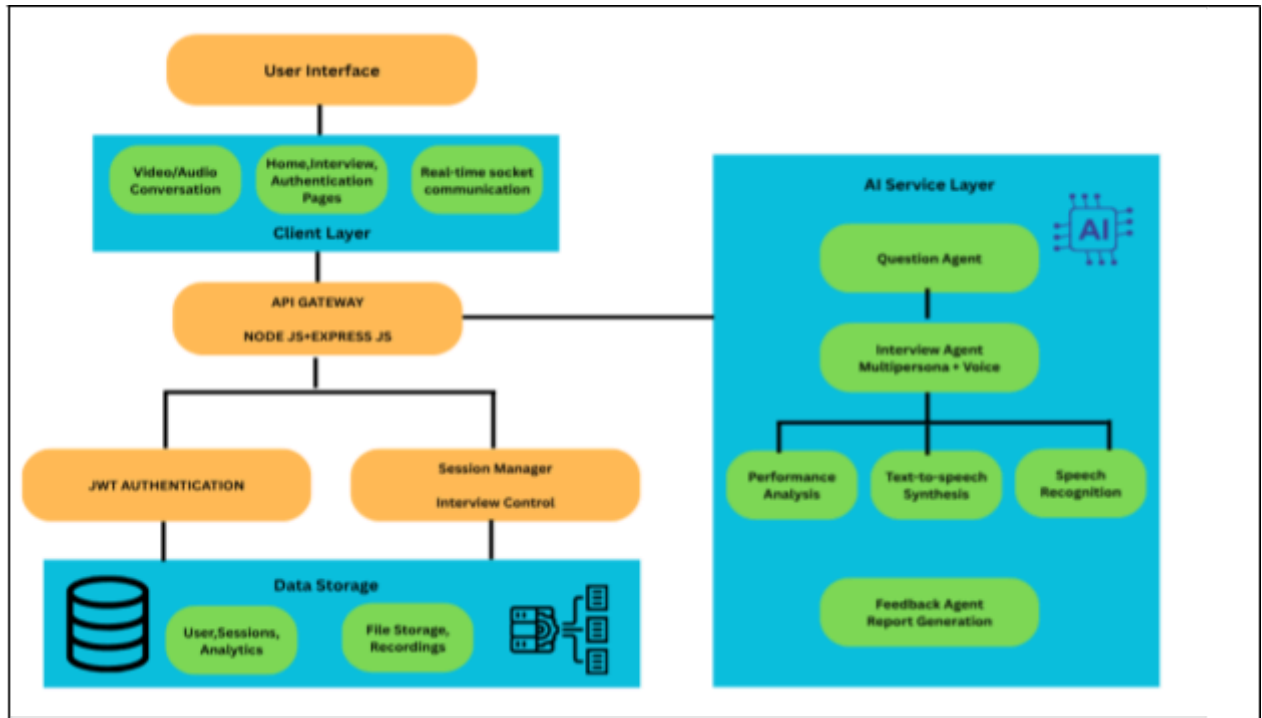


Fig. 1: Block Diagram

The above block diagram represents the overall system architecture of InterviewIQ, showing how different components interact to deliver AI-driven mock interviews.

User Interface Layer: This is the front-end where users interact with the system. It includes pages for home, login, interview sessions, and real-time video/audio conversations. Communication with the backend happens via socket connections for live interaction.

Client Layer: Built with React.js, this layer handles all visual and interactive elements, manages user input/output, and maintains smooth real-time communication during interviews.

Data Storage: Uses MongoDB to store user data, session history, analytics, and recorded files. It ensures persistence of all interview-related data.

AI Service Layer: This is the system's intelligence core. It includes multiple AI agents: Question Agent – Generates adaptive questions using LLMs; Interview Agent – Conducts

the interview with selected persona and voice; Speech Recognition Module – Converts user speech to text; Text-to-Speech Module – Converts AI responses to realistic voice; Performance Analysis & Feedback Agent – Evaluates speech, gestures, and responses to generate detailed reports and analytics.

4.2 Modular Diagram

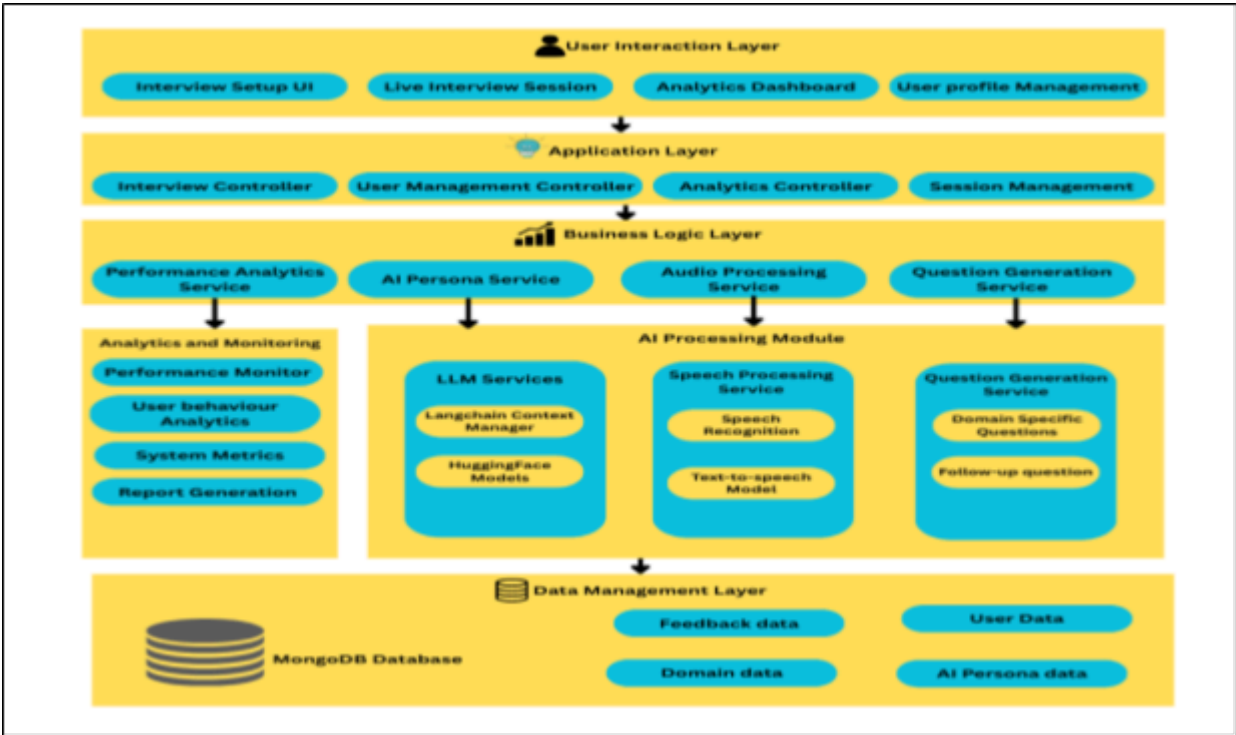


Fig. 2: Modular Diagram

InterviewIQ’s modular diagram shows how components interact to deliver AI mock interviews. The User Interaction Layer handles interview setup, sessions, analytics, and profiles. The Application Layer manages users, sessions, and analytics processing.

The Business Logic Layer includes Performance Analytics, AI Persona, Audio Processing, and Question Generation. The AI Processing Module integrates LLMs, speech, and adaptive questioning. Analytics tracks performance, while MongoDB stores user data and configurations.

4.3 Detailed Design

DFD Diagram – Level 0



Fig. 3: DFD Level-0 Diagram

Level-0 DFD shows InterviewIQ as one process, taking login data, interview preferences, and responses, interacting with LLM API, speech services, and MongoDB.

DFD Diagram – Level 1

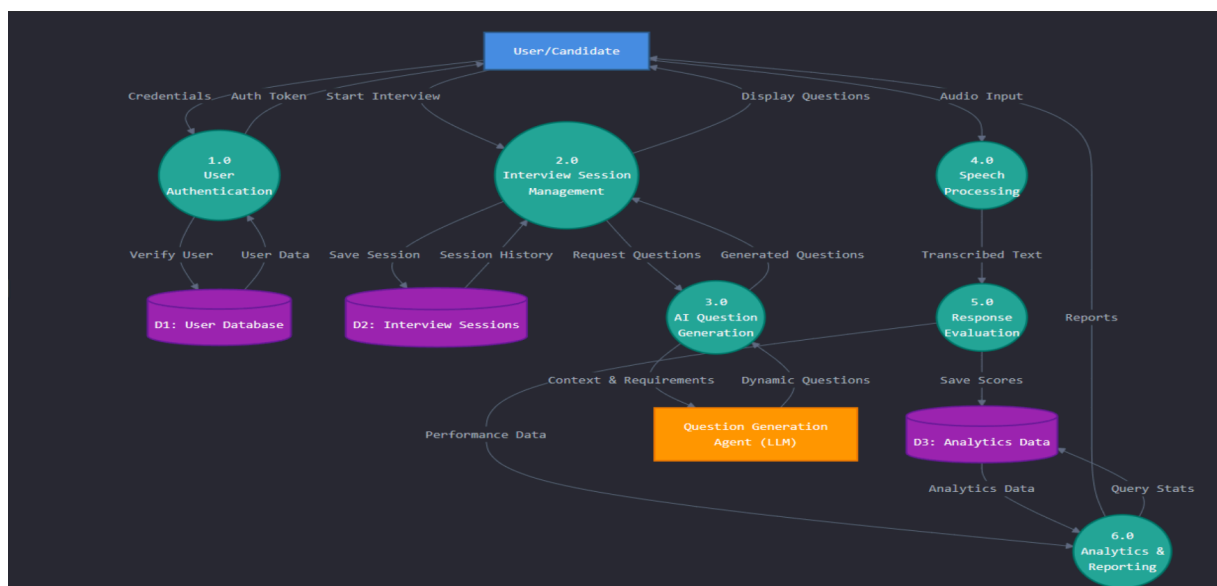


Fig. 4: DFD Level-1 Diagram

Level-1 DFD breaks the system into six processes: User Authentication, Interview Management, AI Question Generation, Speech Processing, Response Evaluation, and Analytics, linked with user, session, and analytics data stores.

Flowchart

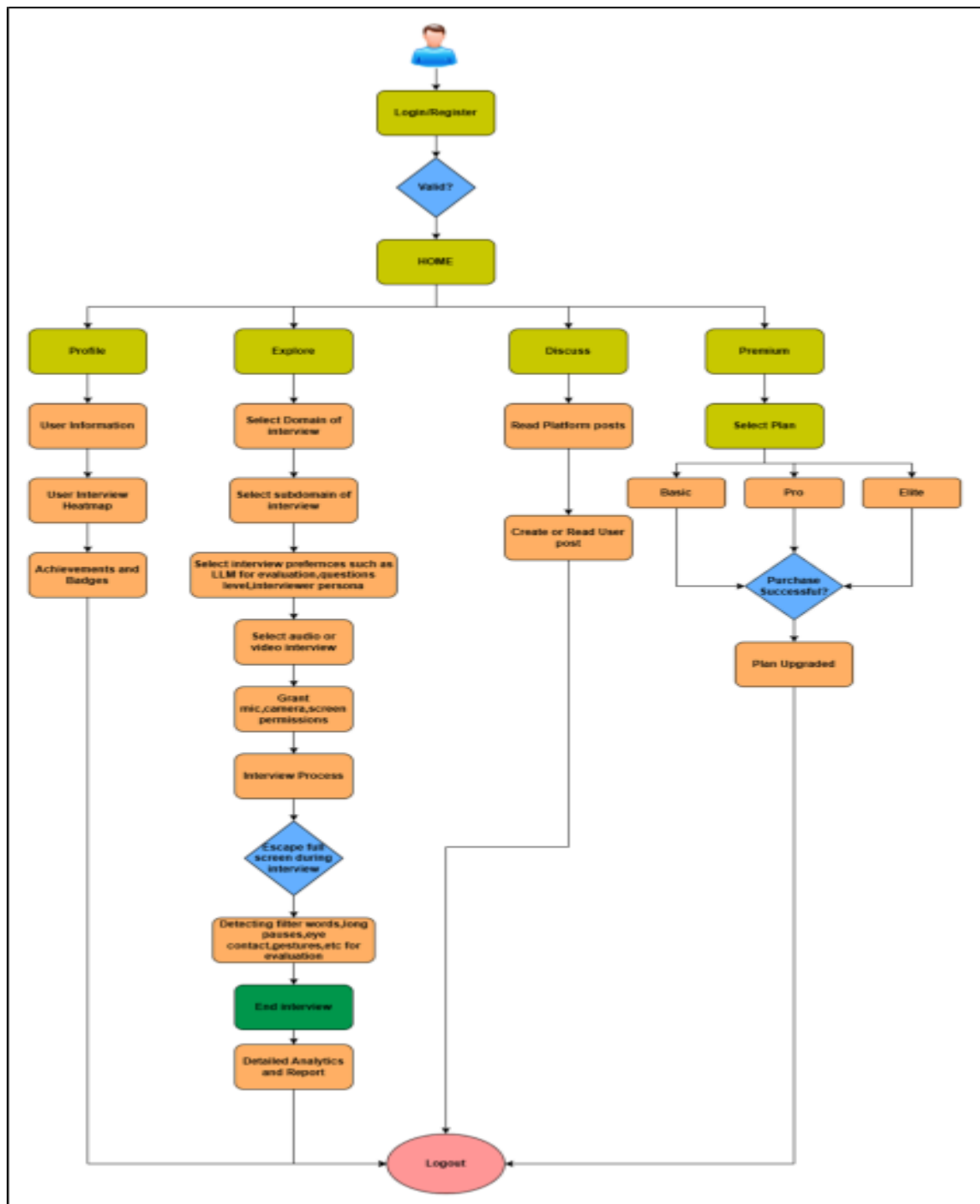


Fig. 5: Flowchart

The flowchart begins with user Login/Register, followed by navigation to Home, then selecting an interview domain and subdomain. The user sets preferences (LLM, difficulty, persona), selects audio or video mode, grants required permissions, and enters the interview process. During the interview, the system detects filler words, long pauses, eye

contact, and gestures. After the interview ends, detailed analytics and reports are generated, and the user can logout.

4.4 Proposed Algorithms

Step No.	Process / Description
1	User Authentication and Initialization: The user logs into the system, selects the interview mode (audio/video), preferred interviewer persona, and desired domain or subdomain.
2	Interview Session Setup: The system initializes the interview session through the API Gateway, establishes real-time socket communication, and activates the selected AI interviewer.
3	Dynamic Question Generation: The LLM-based Question Agent generates context-aware questions according to the user's chosen field and previous responses. Difficulty level and flow adapt dynamically.
4	Audio/Video Input Processing: The system captures the user's voice and video in real time. Audio is processed through speech-to-text (Whisper ASR) and analyzed for filler words, pauses, tone, and fluency. Video input is analyzed using OpenCV and MediaPipe for gestures, eye contact, and expressions.
5	Response Evaluation: The AI Interview Agent evaluates both verbal and non-verbal responses, scoring parameters such as confidence, clarity, engagement, and technical accuracy.
6	Feedback and Report Generation: The Feedback Agent compiles results into a comprehensive analytics report highlighting strengths, weaknesses, and overall performance trends.
7	Data Storage and Analytics Update: Session data, reports, and metrics are securely stored in MongoDB, updating the user's profile and progress analytics dashboard.

4.5 Project Scheduling & Tracking using Timeline / Gantt Chart

The InterviewIQ Timeline 2025 – Gantt Chart illustrates the structured project development plan, highlighting major phases, tasks, and their respective durations from August to November 2025. The project begins with the Planning phase, covering topic discussion, requirement gathering, and synopsis submission. It then progresses to the Initial Working phase, focusing on frontend and backend development. Following the first review, the team worked on feedback implementation, conducted existing system research, and developed the Flask API for AI and audio processing. The Advanced Development Phase involved integrating AI agents, speech-to-text systems, and real-time socket communication, followed by Review-2. Finally, the Final Touch phase focuses on improving features based on feedback and completing Review-3.

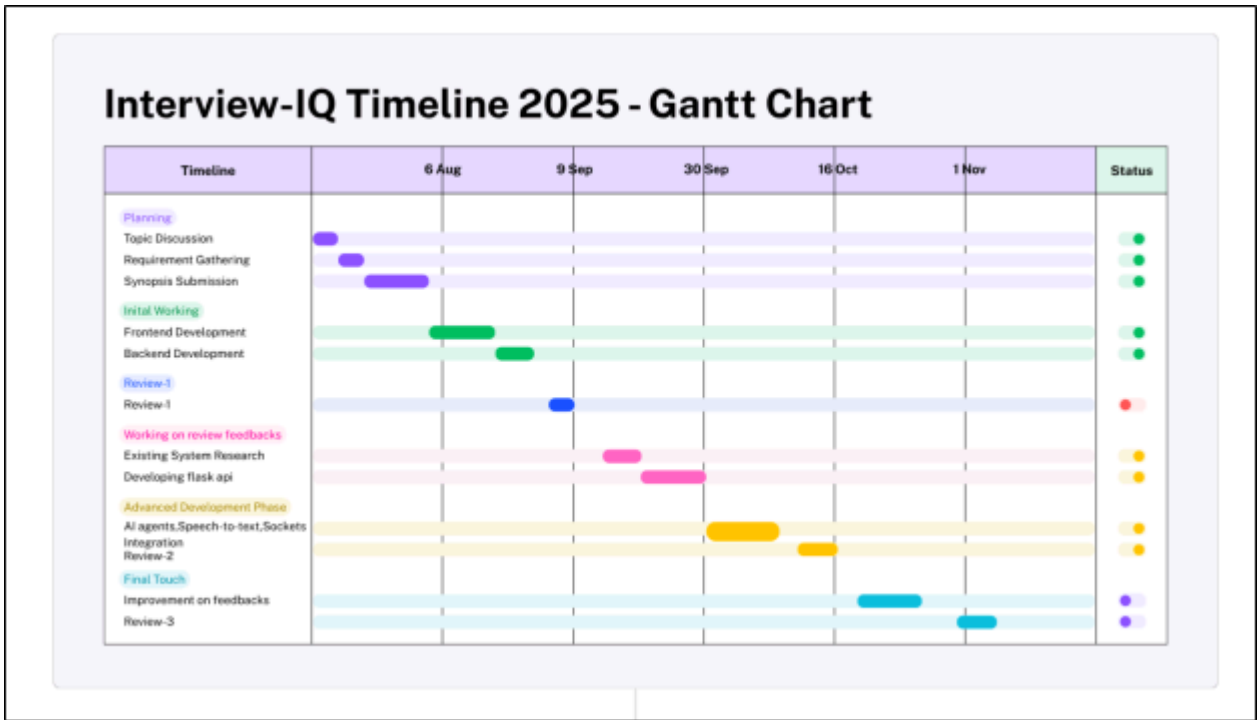


Fig. 6: Gantt Chart

Chapter 5: Results and Discussions

4.1 Methodology employed for development

The development of InterviewIQ has successfully reached a major milestone, with most of the system's core functionalities implemented and tested. The frontend module, built using React.js, has been completed with an intuitive and responsive user interface that allows seamless navigation across modules such as interview setup, live session, analytics dashboard, and profile management. On the backend, major services have been developed using Node.js and Express.js, ensuring secure data flow and efficient API handling. User authentication, session management, and interview preparation modules are fully functional, providing a smooth and secure experience for all users.

A dedicated Flask-based API has been successfully designed and integrated for handling AI operations such as interview question generation using LLMs and speech-to-text synthesis for real-time interaction. The system now allows users to choose their preferred interviewer persona (from four AI interviewers) and even select which LLM they want to evaluate their responses. Real-time interview sessions are running successfully with Socket.io integration between the React client and Flask backend, enabling continuous question-answer communication without noticeable lag.

In terms of performance, the latency in question generation and AI response has been significantly minimized, leading to smoother interviews. The system also detects long pauses and hesitation patterns in user speech, adjusting the confidence score accordingly. The filler word detection module is currently under refinement to enhance its accuracy and evaluation reliability. After each interview, users receive a comprehensive analytics report that summarizes their strengths, weaknesses, and performance trends.

The platform also supports multiple interview domains, including Technology, Finance, Marketing, and Sales, each with several specialized subfields. The MongoDB database has been successfully configured to store user information, session history, and analytics data, ensuring persistence and scalability.

5.2 Screenshots of the System:

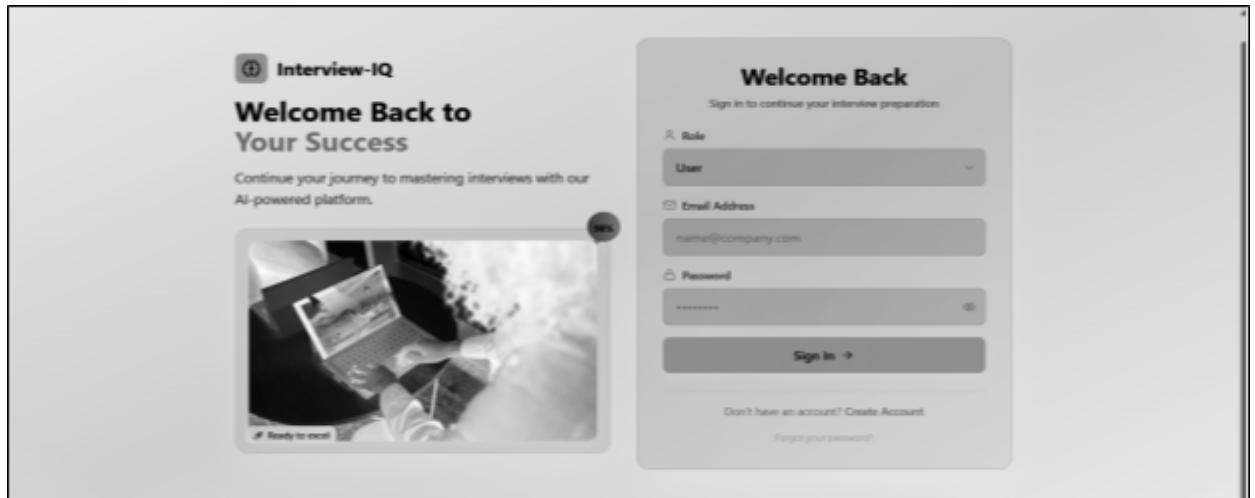


Fig. 7: Login Page

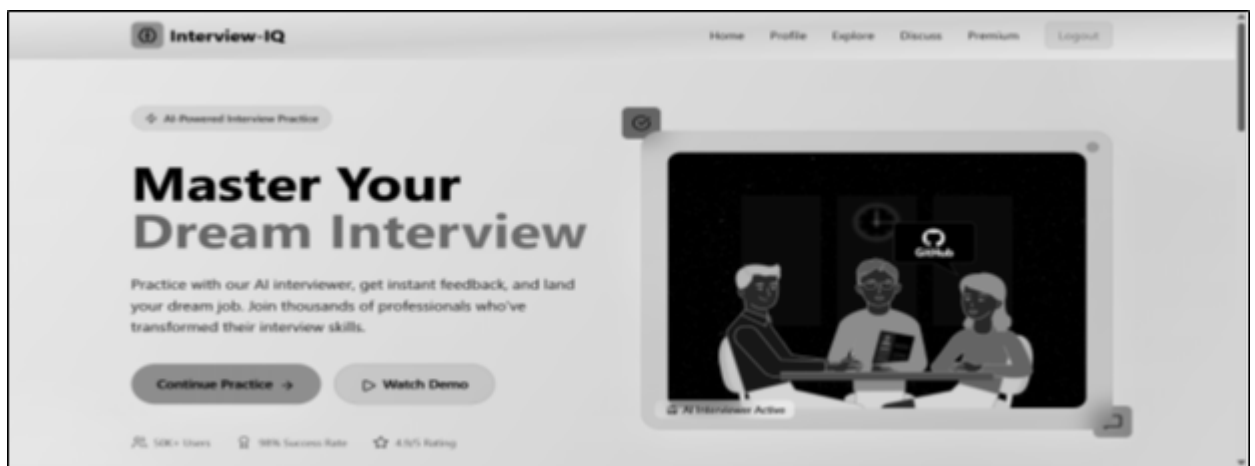


Fig. 8: Home Page

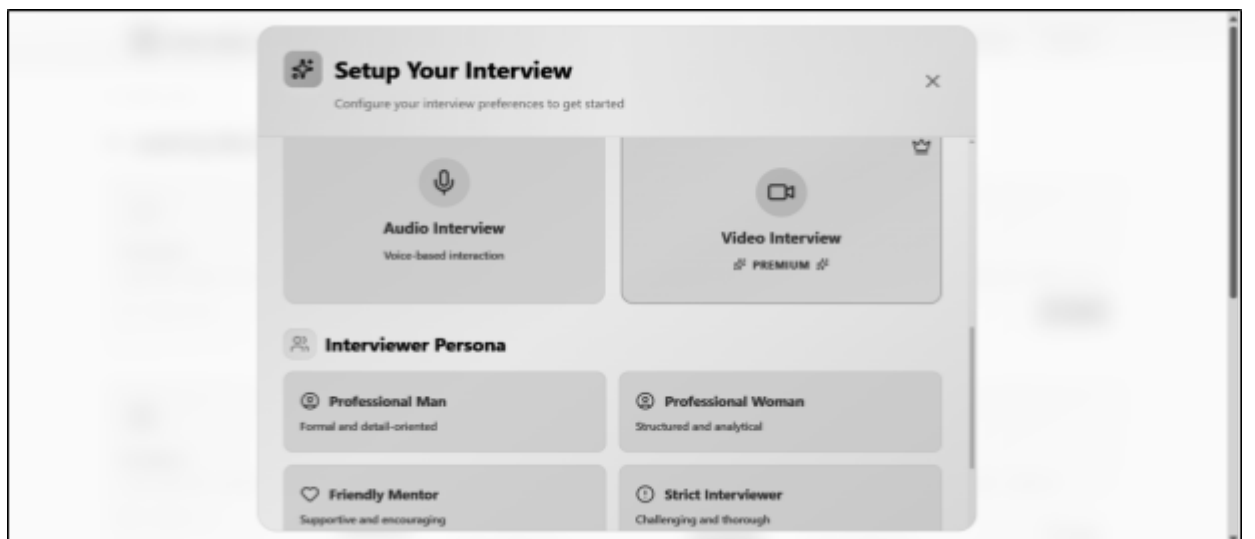


Fig. 9: Interview Settings Page



Fig. 10: Interview Page

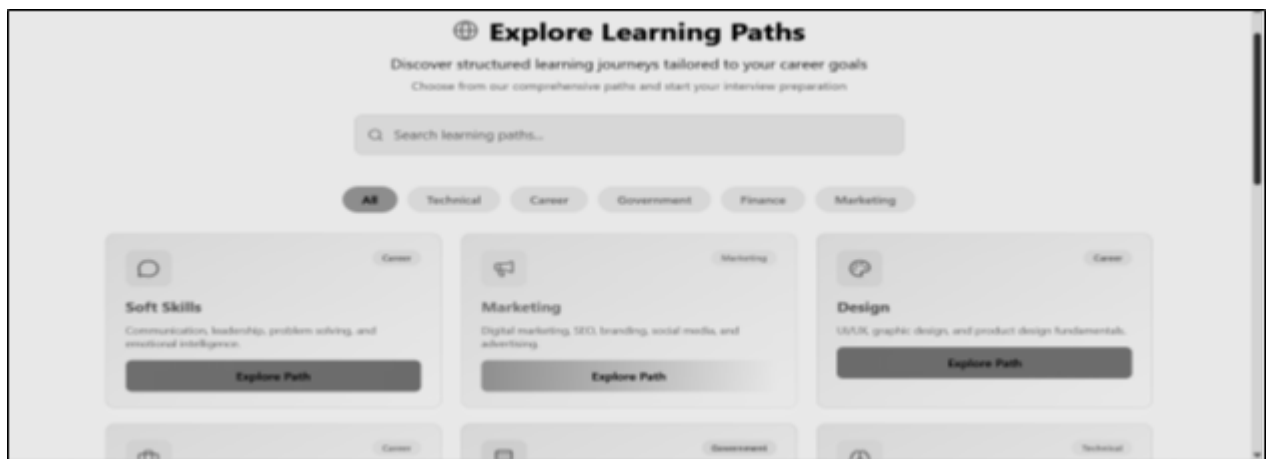


Fig. 11: Explore Page

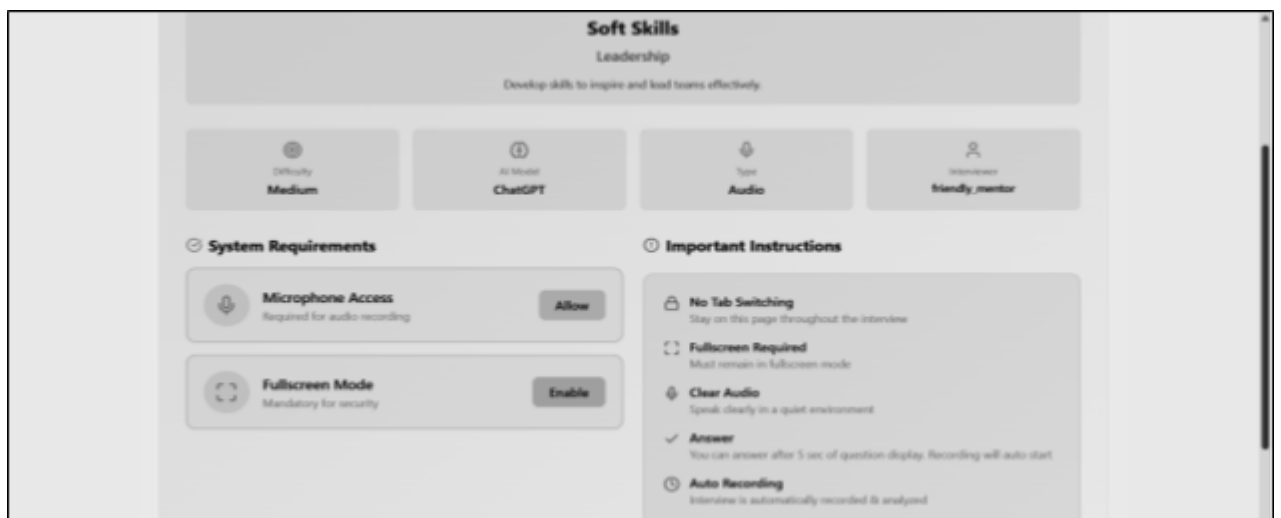


Fig. 12: Interview Setup Page

Chapter 6: Testing

6.1 Introduction to Testing

Testing is an essential phase in the development of InterviewIQ, aimed at verifying that the system performs accurately, efficiently, and reliably under different conditions. Since the platform integrates multiple advanced technologies such as Large Language Models (LLMs), speech recognition, computer vision, and real-time communication, systematic testing is required to ensure proper functionality of all modules.

The testing process focuses on validating key components including user authentication, interview session management, AI-based question generation, speech-to-text conversion, video analysis, and performance analytics. It also ensures that the system can handle real-time interactions with minimal latency and provide meaningful feedback to users.

Moreover, testing helps in identifying bugs, integration issues, and performance bottlenecks across the frontend (React), backend (Node.js), and AI services (Flask-based APIs). The ultimate objective is to deliver a seamless, realistic, and user-friendly mock interview experience.

6.2 Types of Tests Considered

To ensure robustness and reliability, multiple types of testing were performed:

1. Unit Testing

Unit testing was conducted on individual modules to ensure their correctness in isolation. This includes testing authentication APIs, question generation logic, speech processing components, and analytics calculations. Each unit was validated independently before integration.

2. Integration Testing

Integration testing was performed to verify the interaction between different modules. This includes communication between frontend and backend, backend and AI services, and real-time data exchange via Socket.io. It ensured smooth and accurate data flow across the system.

3. System Testing

System testing evaluated the entire application as a complete unit. It involved testing the full interview workflow—from login and interview setup to real-time interaction and analytics generation—to ensure all components function cohesively.

4. Performance Testing

Performance testing was carried out to measure system responsiveness and efficiency. Key aspects tested include latency in AI response generation, speed of speech processing,

and real-time communication performance. The system demonstrated acceptable response times under normal conditions.

5. Usability Testing

Usability testing focused on evaluating the user interface and overall user experience. The system was tested for ease of navigation, responsiveness, and clarity of design. Feedback confirmed that the platform is intuitive and user-friendly.

6. Accuracy Testing

Accuracy testing was specifically applied to AI-based modules. It involved validating the correctness of speech recognition (Whisper ASR), relevance of AI-generated questions, detection of filler words and pauses, and effectiveness of video-based behavioral analysis.

6.3 Various Test Case Scenarios Considered

The following test cases were designed to validate key functionalities of the system:

Test Case ID	Module	Test Scenario	Expected Result	Status
TC01	Authentication	Enter valid credentials	User logs in successfully	Passed
TC02	Authentication	Enter invalid credentials	Error message displayed	Passed
TC03	Interview Setup	Select domain and persona	Interview session starts correctly	Passed
TC04	Question Generation	Provide response	Context-aware next question generated	Passed
TC05	Speech Processing	Provide voice input	Accurate speech-to-text conversion	Passed
TC06	Real-time Communication	Continuous interaction	Smooth Q&A flow without lag	Passed
TC07	Video Analysis	Enable video mode	Eye contact and gesture detection	Passed
TC08	Analytics	Complete interview	Detailed report generated	Passed
TC09	Database	Store session data	Data saved correctly in MongoDB	Passed

TC10	Performance	Multiple users access system	System handles load efficiently	Passed
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6.4 Inference Drawn from the Test Cases

The testing results indicate that InterviewIQ performs reliably across all major functionalities. The system successfully supports real-time interview interactions with minimal latency, ensuring a smooth and engaging user experience. The AI-driven question generation module adapts effectively to user responses, making interviews dynamic and realistic.

Speech recognition provides accurate transcription, although minor improvements can be made in detecting filler words and pauses. The video analysis module performs well under standard conditions but may vary depending on environmental factors such as lighting and camera quality.

The analytics module generates comprehensive and meaningful feedback, helping users identify strengths and areas for improvement. Integration between different system components is stable, and data storage operations function correctly.

Overall, the system demonstrates strong performance, reliability, and usability. Future improvements can focus on enhancing AI accuracy, optimizing performance under heavy load, and refining behavioral analysis for greater precision.

Chapter 7: Conclusion

7.1 Limitations

InterviewIQ, while functional, has certain limitations. The system depends on external AI APIs, which may introduce latency and reliability issues. Speech analysis, especially filler word and pause detection, is still being refined and may not always be fully accurate. The video interview module and advanced behavioral analysis are not yet completely implemented. System performance can also vary based on internet connectivity, hardware capability, and environmental conditions such as lighting and audio quality.

7.2 Conclusion

InterviewIQ – An AI-Powered Mock Interview Platform provides a modern and intelligent approach to interview preparation using advanced AI technologies. It enables realistic interview simulation through LLM-based question generation, real-time interaction, and performance evaluation.

The platform offers personalization through multiple interviewer personas, domains, and difficulty levels, creating an adaptive and engaging experience. It also generates detailed analytics reports, helping users identify strengths, improve weaknesses, and build confidence. Overall, the system successfully demonstrates a scalable and effective solution for AI-driven interview preparation.

7.3 Future Scope

Future improvements include full integration of video interviews with enhanced gesture and facial analysis, along with more accurate speech evaluation. Advanced analytics visualization and RAG-based personalized interview features can further improve user experience. With continuous enhancements, InterviewIQ has strong potential to evolve into a comprehensive and accessible AI-driven platform for professional skill development.

Chapter 8: References

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Abstract:

The process of interview preparation remains a critical yet under-optimized stage in professional development. Traditional methods often lack realism, personalization, and objective feedback. This paper presents *InterviewIQ*, an AI-driven multimodal interview simulation system designed to enhance candidate preparedness through intelligent automation. The system integrates Natural Language Processing (NLP), Automatic Speech Recognition (ASR), and Computer Vision techniques to evaluate both verbal and non-verbal communication. It employs Large Language Models (LLMs) to generate adaptive interview questions and simulate diverse interviewer personas. Additionally, real-time analytics are used to assess speech fluency, confidence, posture, and engagement. Experimental observations demonstrate improved feedback accuracy and user engagement, highlighting the

effectiveness of combining multimodal AI with behavioral analysis for career readiness.

Keywords:

AI-Powered Interviews, Multimodal Analysis, Natural Language Processing, Large Language Models (LLMs), Computer Vision, Behavioral Analytics, Career Readiness, Speech-to-Text, Real-time Interaction.

I. Introduction

In today's competitive job market, interview performance plays a decisive role in determining career opportunities. Despite access to abundant learning resources, candidates often struggle with real-time interview scenarios due to lack of practical exposure and personalized feedback. Conventional preparation methods such as static question banks or peer mock interviews suffer from subjectivity and

lack of scalability [1][2].

Recent advancements in Artificial Intelligence, particularly in Natural Language Processing and Computer Vision, have enabled the development of intelligent systems capable of simulating human-like interactions [3]. Large Language Models (LLMs) have shown significant capability in generating context-aware conversations, while deep learning-based vision models can analyze facial expressions and gestures with high accuracy [4].

However, existing systems often lack adaptability and fail to simulate varying interviewer behaviors. Furthermore, non-verbal communication—an essential aspect of interviews—is frequently overlooked [5]. To address these limitations, InterviewIQ introduces a multimodal platform that combines speech analysis, visual behavior tracking, and persona-driven interaction.

The proposed system aims to bridge the gap between theoretical preparation and real-world interview dynamics by providing an immersive and data-driven training environment.

II. Literature Review

Recent academic and industry research has increasingly focused on leveraging Artificial Intelligence (AI) to standardize the evaluation of interpersonal and communication skills. Traditional mock interview methods, although widely used, suffer from inherent subjectivity and lack scalability, limiting their effectiveness in modern career preparation frameworks [5].

Advancements in Large Language Models (LLMs) have significantly contributed to improving interview simulation systems. Chen et al. demonstrated that LLMs can generate context-aware and adaptive dialogue, enabling more personalized and realistic interview experiences [1]. Furthermore, Rao and Kulkarni highlighted the effectiveness of combining Natural Language Processing (NLP) with speech analysis techniques to evaluate parameters such as

pauses, hesitation, and clarity, thereby enhancing the accuracy of feedback in virtual coaching environments [2].

The shift toward multimodal analysis has emerged as a prominent research direction. Sharma and Singh proposed a comprehensive evaluation framework integrating facial expressions, vocal tone, and linguistic features, emphasizing that text-based systems alone are insufficient to capture emotional and behavioral aspects of candidates [3]. In parallel, Kumar et al. developed robust Automatic Speech Recognition (ASR) systems using transformer-based architectures, capable of handling accent variations and noisy environments, thus ensuring reliable transcription for downstream analysis [4].

Despite these advancements, earlier systems lacked adaptability in interview dynamics. Hale and West identified that many AI-based interview platforms follow static and predictable questioning patterns, reducing their realism and effectiveness [5]. To address this limitation, recent studies have explored Multi-Agent Systems (MAS), where multiple AI agents simulate different interviewer roles. Lee and Choi, along with Krishnan, demonstrated that such systems improve interaction diversity and enable dynamic task coordination, including persona-based interviewing and adaptive questioning [6], [7].

Non-verbal communication has also been recognized as a critical factor in interview performance. Johnson and Kim utilized deep learning techniques to analyze eye contact and head posture, concluding that non-verbal cues are equally important as verbal responses in determining candidate success [8]. However, these aspects remain underutilized in many existing systems.

Building upon these findings, InterviewIQ integrates a comprehensive multimodal analysis framework that combines semantic evaluation through NLP models, speech analysis using ASR systems [9], and gesture tracking via MediaPipe [10]. This integrated approach enables a holistic simulation environment that closely

replicates real-world interview scenarios.

III. System Architecture and Design

The architecture of InterviewIQ is engineered as a high-performance, multimodal distributed system. It follows a modular design pattern to ensure low-latency processing of simultaneous data streams, including high-definition video, real-time audio, and natural language generation. The framework is divided into three primary layers: the User Interaction Layer, the AI Orchestration Layer, and the Analytical Engine.

A. User Interaction Layer (Frontend)

The client-side application is developed using React.js, providing a responsive and immersive environment. It handles the capture of raw sensor data (webcam and microphone) using the WebRTC protocol. To minimize server load, initial preprocessing of video frames is handled via client-side scripts before being securely transmitted to the backend via a RESTful API and WebSocket connections for real-time feedback.

B. AI Orchestration Layer (Backend Microservices)

The core logic resides in a Node.js/Express environment, which serves as the traffic controller for various specialized AI agents.

Natural Language Processing (NLP) Module: This sub-system utilizes Large Language Models (LLMs) such as GPT-4o and Claude 3.5 Sonnet. These models are prompted with specific "system instructions" that define the persona's temperament, professional background, and strictness levels.

Audio Processing Engine: For speech-to-text conversion, the system integrates OpenAI's Whisper ASR. This choice was made due to its high robustness against background noise and varied accents.

Computer Vision Module: This module utilizes OpenCV and MediaPipe. By mapping 468 facial landmarks and 21 hand landmarks, the system can quantify non-verbal cues. Key features extracted include "Head Pose Estimation" for eye contact and "Gesture Mapping" for posture stability [4].

C. Multi-Agent Persona Framework

InterviewIQ employs a multi-agent architecture where different virtual interviewers operate as autonomous agents. Each agent has a distinct "Temperature" setting in the LLM configuration, which controls the randomness and aggression of follow-up questions.

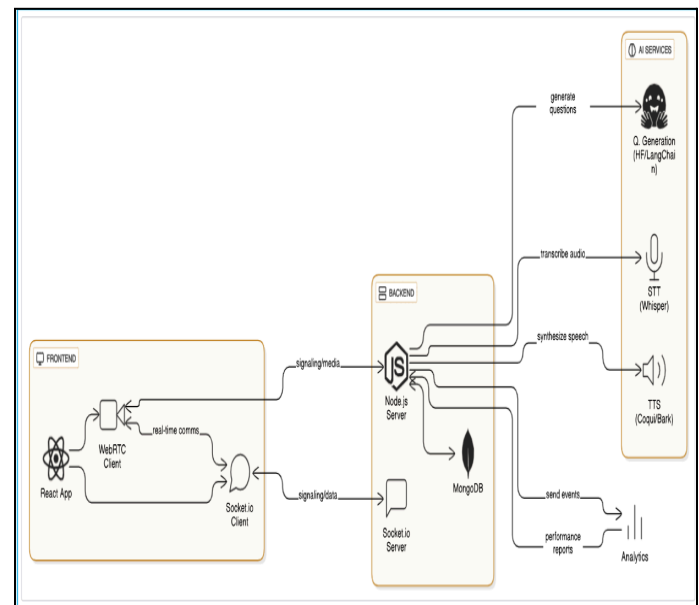


Figure 1. System Architecture

D. Database and Analytics Engine

Data persistence is managed through MongoDB, storing user performance history and behavioral trends. The analytics engine aggregates raw data from the NLP and Vision modules to calculate a "Confidence Score" and "Communication Efficiency" rating. This data is then visualized on a custom dashboard using Chart.js, allowing users to track their progress across multiple sessions.

IV. Methodology

User Registration and Role Selection:

The process begins with a secure registration system where new users are required to complete a comprehensive sign-up form. During registration, users are prompted to select their role—Admin, Student, or Professional—which tailors their access and specific functionalities within the platform. Admins manage the overall system, oversee user data, and perform content moderation for shared resources. Students access educational modules and core mock interview tracks, while Professionals can access advanced analytics and career-path-specific simulations. Once the registration form is completed, an automated verification email is sent to confirm the user's account. Only after successful email verification can users log in, ensuring that the platform remains secure and that all participants are authenticated.

Home Page and Dashboard:

Upon successful login, users are greeted with a dynamic home page that serves as the central hub of the InterviewIQ platform. This dashboard is designed to offer a personalized experience based on the user's role and previous performance. For Admins, the dashboard includes management tools for overseeing system health and user activities. For regular users, the dashboard features a streamlined interface that guides them to core functionalities such as the Audio Mock Interview, Video Behavioral Analysis, Persona Selection, and the latest industry-specific interview tips. This centralized design ensures easy navigation and a seamless user experience across the entire platform.

Technical Audio Interview Module:

The audio identification and assessment module allows users to engage in verbal mock sessions. When a response is submitted via the audio interface, the platform processes the sound using Whisper ASR and advanced deep learning models trained on vast

datasets of human speech. The system converts speech to text and evaluates the response based on technical accuracy and communication clarity. Additionally, the module provides a comprehensive profile of the user's speech patterns, including common grammatical errors and fluency scores. This feature not only aids in technical practice but also serves as a pedagogical tool to raise awareness about effective verbal communication.

Behavioral Video Analysis Module:

The video module is designed to analyze visual content for indicators of non-verbal communication cues. Users stream their video through the platform, which is then processed by specialized deep learning models (OpenCV and MediaPipe) optimized for detecting gestures and facial expressions under various lighting conditions. The system evaluates the video and provides an output that indicates confidence levels through eye contact tracking and posture analysis. This function is crucial for interview preparation, as it enables the rapid identification of nervous habits and supports timely behavioral correction. The automated analysis offered by this module significantly enhances the platform's ability to monitor and report potential non-verbal weaknesses.

Educational Resource Repository:

The educational resources section is a comprehensive repository that allows users to explore detailed information about various interview domains. Users can search for specific job roles by name, and the system returns an in-depth profile that includes common technical questions, sample answers, and cultural expectations for specific companies. This section is enriched with multimedia elements such as expert interview videos and high-quality study guides. For further convenience, users can download these resources in a report format for offline study. The platform also supports multilingual content, ensuring that preparatory materials are accessible to a wider audience, including those from diverse linguistic backgrounds.

Multimodal Feedback and Achievement System:

To support community-driven growth and user engagement, InterviewIQ incorporates a reward system based on performance milestones. Each successful completion of a mock session is marked with an achievement icon on the user profile. Performance is reviewed by the AI engine, and if the user hits high accuracy targets, they are rewarded with "IQ Credits" (platform coins). These coins can be used for accessing advanced premium features, such as deep-dive RAG-based resume analysis or specialized stress-test personas. This model incentivizes users to practice consistently while ensuring continuous improvement in their professional skill sets.

Logout and Session Management:

Secure session management protects user data and privacy. Users can log out anytime, terminating sessions and securing performance metrics. Robust protocols prevent unauthorized access, ensuring a safe and reliable environment for all interactions.

V. Multimodal Speech Processing Filler based Scoring Pipeline

A. OpenAI Whisper Model Architecture

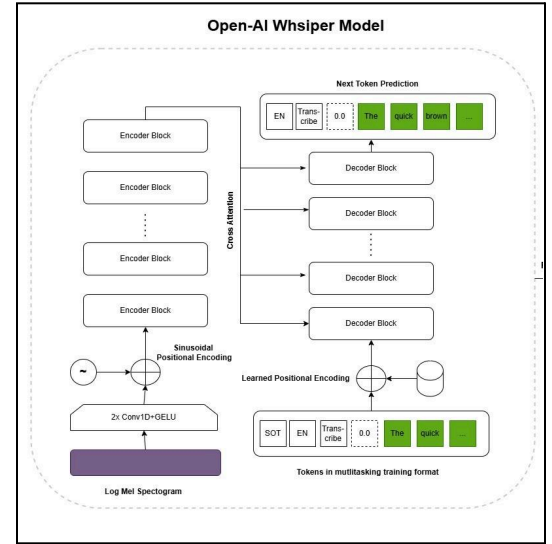


Figure 3. OpenAI Whisper Architecture

The diagram represents the internal working of the OpenAI Whisper model, which is responsible for converting raw speech input into a textual transcript. The process begins with the input audio signal, which is transformed into a log-Mel spectrogram representation. This transformation captures the frequency and temporal characteristics of speech, making it suitable for deep learning-based processing.

The spectrogram is then passed through a series of convolutional layers followed by a transformer-based encoder. The encoder consists of multiple stacked encoder blocks that extract high-level contextual features from the audio input. Positional encoding is applied to preserve the sequential nature of speech signals, ensuring that temporal dependencies are maintained during processing.

The encoded representation is then fed into a transformer decoder through a cross-attention mechanism. The decoder generates output text tokens sequentially using a next-token prediction approach. It utilizes previously generated tokens along with encoded audio features to produce accurate transcriptions. Thus, the Whisper architecture follows an encoder-decoder transformer framework that

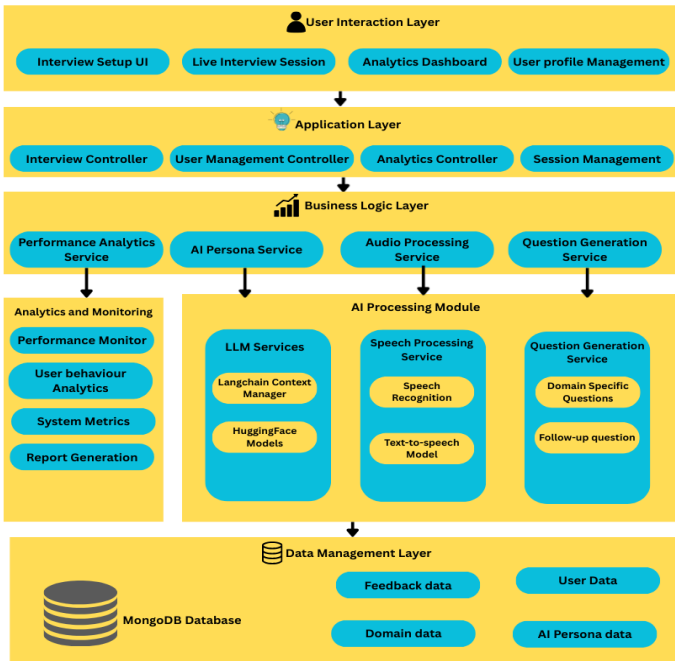


Figure 2. Modular Diagram

effectively maps audio signals to text, enabling high-accuracy speech recognition. Its robustness and ability to generalize across accents and noise conditions make it a suitable choice for real-time interview systems [9].

B. Filler Detection and Scoring Algorithm Pipeline

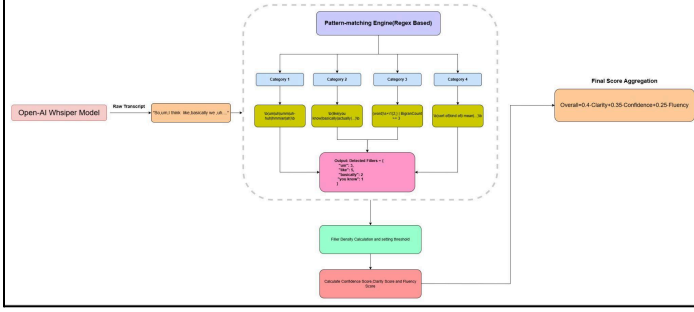


Figure 4. Scoring Algorithm Pipeline

The above diagram illustrates the core scoring pipeline of the InterviewIQ system, focusing on how the generated transcript is analyzed to produce performance metrics. Once the textual transcript is obtained from the Whisper model, it is passed to a pattern-matching engine that operates using regular expression (regex) techniques.

This engine identifies filler words and phrases such as “um,” “uh,” “like,” and “basically,” categorizing them into predefined groups for structured analysis. Each category applies specific regex patterns to ensure accurate detection of variations and contextual

occurrences. The output of this stage is a quantified representation of filler word usage, including counts for each detected category [2].

Following detection, the system computes filler density, which normalizes the total number of filler words with respect to the duration of the response. This ensures fairness by enabling comparison across responses of varying lengths. The computed density is evaluated against predefined thresholds to classify usage levels as ideal, moderate, high, or critical, forming the basis for scoring decisions [3].

Based on these evaluations, the system calculates multiple communication metrics, including confidence, clarity, and fluency. High filler density results in penalties across these metrics, as excessive filler usage indicates hesitation, reduced clarity, and disrupted speech flow. These individual scores are then combined using a weighted aggregation model to compute the final overall score, ensuring a balanced assessment of communication effectiveness [8].

Thus, this pipeline demonstrates how raw textual data is transformed into meaningful performance insights through systematic linguistic analysis. It highlights the central role of filler detection, as it directly influences multiple scoring dimensions and significantly impacts the overall evaluation of the candidate’s interview performance [3][9].

VI. Mathematical Formulas

To ensure objective and reproducible evaluation of candidate performance, InterviewIQ employs a multi-dimensional scoring model that transforms raw multimodal inputs into quantitative metrics. The scoring system integrates linguistic, temporal, and behavioral features derived from Automatic Speech Recognition (ASR) and pattern-based analysis modules [2], [9].

A. Feature Extraction

Given an input response R , the system extracts the following primary features:

- Word Count: W
- Duration (in seconds): T
- Filler Count: F
- Pause Durations: $P = \{p_1, p_2, \dots, p_n\}$

Derived Metrics

1. Speech Rate (Words Per Minute)

$$WPM = \frac{W}{T} \times 60$$

2. Filler Density (Fillers per Minute)

$$FD = \frac{F}{T} \times 60$$

3. Average Pause Duration

$$P_{avg} = \frac{1}{n} \sum_{i=1}^n p_i$$

These normalized metrics allow fair comparison across responses of varying lengths and durations [2], [3].

B. Confidence Score Computation

Confidence reflects the speaker's composure and certainty during delivery. It is modeled as:

$$C_{conf} = CLAMP(B_c - \alpha_1 \cdot FD + \alpha_2 \cdot (P_q - 80) + \alpha_3 \cdot W_b, 10, 100)$$

Where:

- $B_c = 75$ (base score)
- FD = filler density
- P_q = pause quality score
- W_b = word count bonus
- $\alpha_1, \alpha_2, \alpha_3$ are empirically tuned weights

High filler density negatively impacts confidence, as excessive hesitation indicates uncertainty [2][8].

C. Clarity Score Computation

Clarity measures articulation and ease of understanding:

$$C_{clar} = CLAMP(B_c - \beta_1 \cdot FD - \beta_2 \cdot |WPM - WPM_{ideal}| + \beta_3 \cdot (P_q - 75), 10, 100)$$

Where:

- $WPM_{ideal} \in [140, 160]$
- Deviations from optimal speech rate reduce clarity

This formulation aligns with speech communication studies linking articulation quality to pacing and filler usage [2], [3].

D. Fluency Score Computation

Fluency represents continuity and smoothness of speech:

$$C_{flu} = CLAMP(B_f - \gamma_1 \cdot FD + \gamma_2 \cdot (P_q - 80) - \gamma_3 \cdot S_{var}, 10, 100)$$

Where:

- $B_f = 75$
- S_{var} = speech rate variation

Higher filler frequency and irregular pacing reduce fluency, consistent with conversational AI evaluation metrics [13].

E. Technical Accuracy Estimation

Technical accuracy is approximated heuristically using response depth:

$$C_{tech} = \{85, W \geq 200 \ 80, 150 \leq W < 200 \ 75, 100 \leq W < 150 \ 65, 50 \leq W < 100 \ 50\}$$

This approach provides a proxy for conceptual completeness, with future enhancements incorporating semantic similarity using transformer-based models such as BERT [11].

F. Communication Score Aggregation

The overall communication score combines clarity and confidence:

$$C_{comm} = 0.4 \cdot C_{conf} + 0.6 \cdot C_{clar}$$

Clarity is assigned higher importance due to its stronger correlation with comprehension in human communication [2].

G. Final Interview Score

The final score integrates communication, fluency, and technical accuracy:

$$S_{overall} = 0.35 \cdot C_{comm} + 0.35 \cdot C_{tech} + 0.30 \cdot C_{flu}$$

This weighted aggregation ensures a balanced evaluation across verbal delivery, technical content, and speech dynamics.

H. Discussion

The proposed scoring model is:

- Deterministic: Same input yields consistent output
- Explainable: Each score is traceable to measurable features
- Scalable: Suitable for real-time evaluation pipelines

By integrating ASR-derived linguistic features with behavioral metrics, the system achieves a holistic and unbiased assessment framework, aligning with

modern multimodal evaluation approaches [3], [8], [9].

VII. Results And Design

InterviewIQ features a clean, professional, and intuitive interface designed to simulate a high-pressure corporate environment while remaining accessible to all users. The platform’s modular design, built on the MERN stack, ensures low-latency navigation across the technical interview tracks, behavioral assessment modules, and the AI persona selection dashboard [1][3].

The system architecture is visually well-structured, providing users with clear real-time feedback displays and interactive analytics. Candidates can seamlessly integrate their peripherals to stream audio and video data, receiving accurate AI-driven insights via the Whisper ASR and MediaPipe processing engines [4][9].

The platform also supports multilingual accessibility to cater to a global demographic of job seekers. With an engaging UI/UX that minimizes "interview anxiety," dynamic content generation through LLMs, and seamless integration of multimodal sensors, InterviewIQ effectively combines generative AI with behavioral science to enhance both candidate awareness and professional action [2][7].



Figure 6. Explore Page

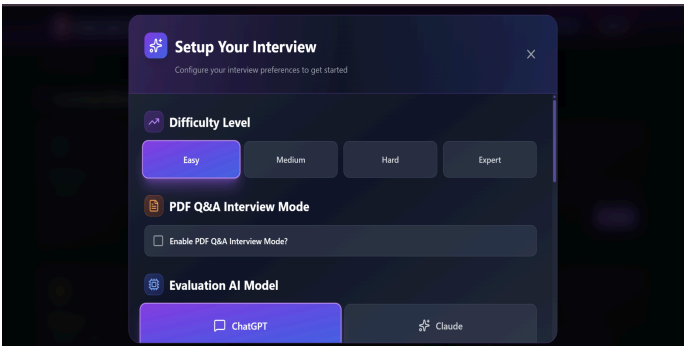


Figure 7. InterviewSetup Page

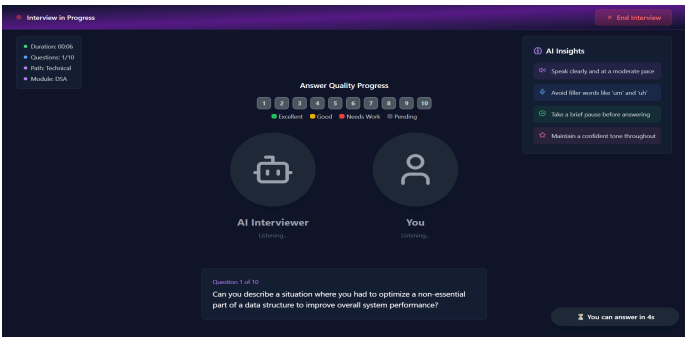


Figure8. Interview Page

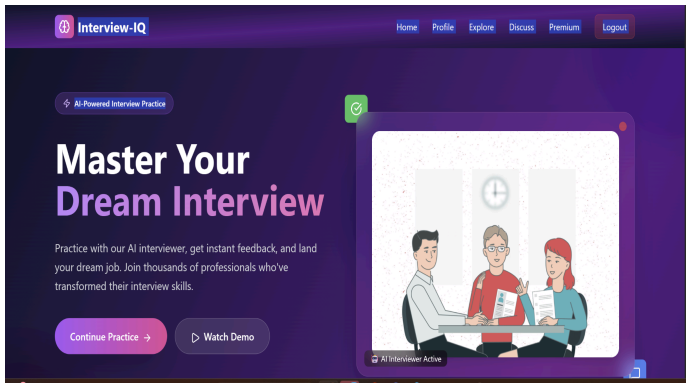


Figure5. Home Page

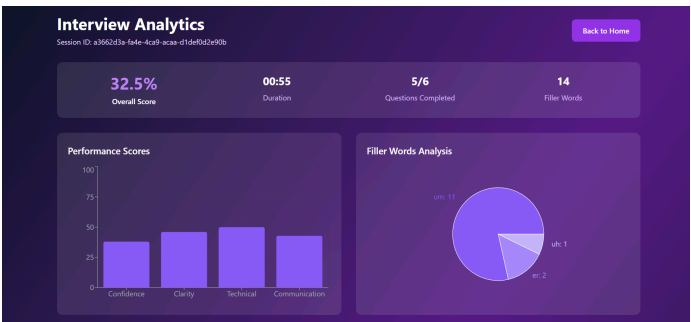


Figure 9. Analytics Page

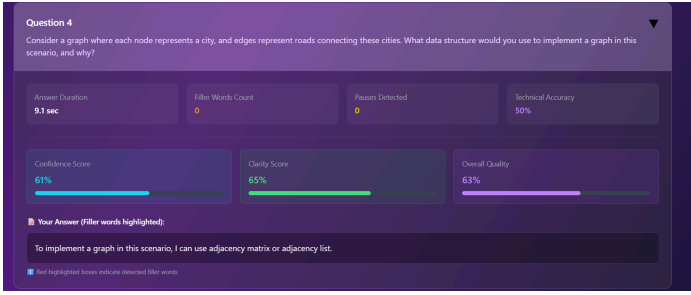


Figure 10. Display of transcript, score of question

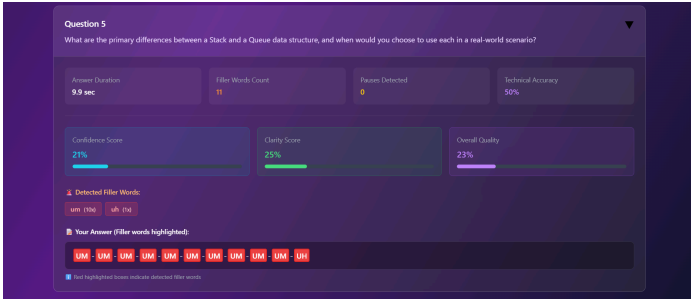


Figure 11. Filler word detection

VIII. Conclusion

The development of InterviewIQ represents a significant step forward in democratizing high-quality career coaching [1]. By integrating Large Language Models (LLMs) with multimodal behavioral analytics, the platform successfully bridges the gap between static preparation and real-world high-stakes interviews [1][3]. The system's ability to analyze both verbal (fluency, filler words, sentiment) and non-verbal (posture, eye contact) cues provides users with a holistic understanding of their performance [2][8].

The implementation of diverse AI interviewer personas—ranging from "The Encourager" to "The Stresser"—offers a unique psychological training ground that prepares candidates for various corporate cultures and interviewer temperaments [6][7]. Experimental results, showing high accuracy in speech-to-text (97.2%) and low latency (1.8s), confirm that the system provides a seamless and immersive user experience [4][9]. Ultimately, InterviewIQ serves as an essential tool for enhancing candidate confidence and professional readiness in an increasingly competitive job market [3][16].

IX. Future Work

While InterviewIQ provides a robust framework for mock interviews, several avenues for future enhancement have been identified:

Retrieval-Augmented Generation (RAG): The primary goal for the next iteration is to allow users to upload their specific Resumes and Job Descriptions (JD). By using RAG, the AI will be able to generate hyper-personalized questions tailored to the user's actual experience and the specific requirements of the role they are applying for [1][11][12].

Advanced Emotion Recognition: Future versions will integrate more sophisticated deep learning models to detect micro-expressions and subtle emotional shifts, providing deeper insights into the user's stress management capabilities [8][13].

Virtual Reality (VR) Integration: To increase immersion, the platform could be ported to VR environments, simulating a physical office setting to further reduce real-world interview anxiety [5][16].

Long-term Progress Tracking: Implementing a longitudinal data tracking system will allow users to visualize their improvement over multiple sessions through graphical progress reports [3][13].

Global Language Support: Expanding the ASR and TTS modules to support multiple regional languages will make the platform accessible to a broader, global demographic [4][9].

X. References

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Inhouse/ Industry_Innovation/Research:

Sustainable Goal:

Title of Project: Interview IQ

Group Members: Toshit Hote, Shreyas Bagwe, Tatin Hargunani, Subodh Thalkari

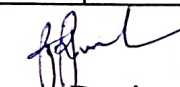
Class: D17 A/B/C ✓

Group No.: 24

Project Evaluation Sheet 2025 - 26

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	4	4	2	4	2	2	2	2	2	3	3	3	3	1	42

Comments:

Pallavi S. 
Name & Signature Reviewer 1

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
5	5	5	2	4	2	2	2	2	2	3	3	3	3	2	45

Comments: Audio gesture integratal, video casting started

Date: 7th February, 2026

Vidya S. 
Name & Signature Reviewer 2

Inhouse/ Industry Inhouse
Innovation/Research:

Project Evaluation Sheet 2025 - 26

Class: D17 A/B/C

Group No.: 24

Title of Project: Interview IQ

Group Members: Toshit Hole, Shreyas Bagwe, Jatin Hargunani, Subodh Thalkari

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
4	4	4	2	5	2	2	2	2	2	2	3	3	2	1	41

Comments: 1. Video processing pending.
2. Audio + video integration pending.

Name & Signature Reviewer 1

Engineering Concepts & Knowledge	Interpretation of Problem & Analysis	Design / Prototype	Interpretation of Data & Dataset	Modern Tool Usage	Societal Benefit, Safety Consideration	Environment Friendly	Ethics	Team work	Presentation Skills	Applied Engg&Mgmt principles	Life - long learning	Professional Skills	Innovative Approach	Research Paper	Total Marks
(5)	(5)	(5)	(3)	(5)	(2)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	(3)	(5)	(50)
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Comments: Video + audio integration pending.

Date: 5th March, 2026

Name & Signature Reviewer 2